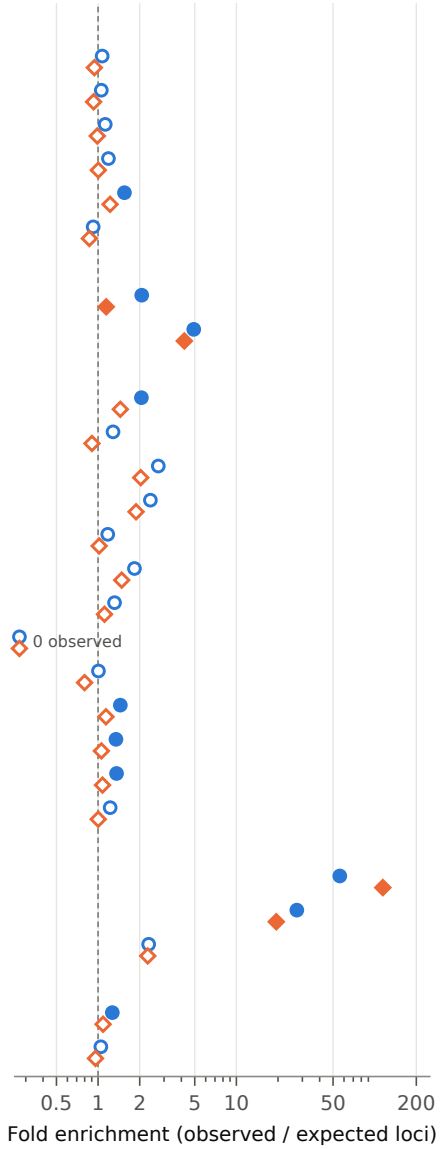
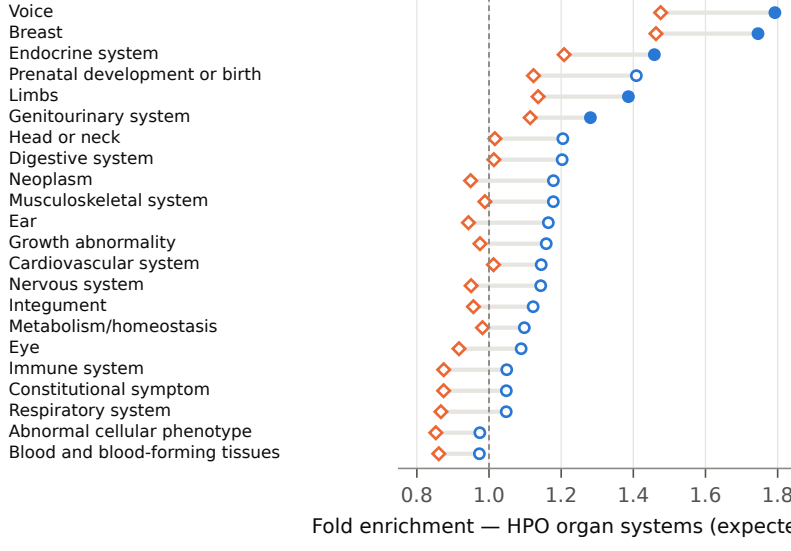
A

	obs / exp loci	p, primary null	p, constraint-matched null: gnomAD v2.1.1	p, constraint-matched null: gnomAD v4.1
Mendelian disease genes				
Any Mendelian (OMIM or Orphanet)	123 / 114.9	0.16	0.87	0.69
OMIM	118 / 111.8	0.22	0.91	0.77
Orphanet (assessed)	111 / 98.5	0.054	0.61	0.38
ClinGen moderate–definitive	77 / 64.8	0.041	0.51	0.29
Autosomal dominant	80 / 51.6	1.1×10^{-5}	0.018	0.011
Autosomal recessive	74 / 80.3	0.84	0.96	0.97
Controls				
LoF-constrained decile (gnomAD)	107 / 51.7	7.0×10^{-16}	0.0095	3.8×10^{-6}
GO: circadian rhythm	22 / 4.5	8.4×10^{-10}	7.7×10^{-9}	1.5×10^{-8}
Pre-specified phenotypes				
Sleep disturbance	34 / 16.5	3.5×10^{-5}	0.013	0.0039
without neurodevelopmental features	4 / 3.1	0.38	0.66	0.61
Sleep–wake cycle / melatonin	3 / 1.1	0.098	0.18	0.12
Prolonged QT interval	3 / 1.3	0.13	0.21	0.17
Arrhythmia	17 / 14.5	0.28	0.51	0.38
Hypertrophic cardiomyopathy	13 / 7.1	0.024	0.094	0.054
Cardiomyopathy	21 / 16.0	0.11	0.33	0.22
Reduced circulating complement	0 / 0.90	1.00	1.00	1.00
Immunodeficiency	5 / 5.0	0.56	0.76	0.69
Global developmental delay	76 / 52.5	1.7×10^{-4}	0.086	0.036
Intellectual disability	71 / 52.7	0.0029	0.30	0.15
Seizure	70 / 51.5	0.0022	0.24	0.12
Short stature	44 / 36.0	0.082	0.52	0.38
Post hoc				
Monoamine neurotransmitter disorders	4 / 0.07	6.7×10^{-7}	4.0×10^{-8}	4.1×10^{-8}
Decreased CSF homovanillic acid	3 / 0.11	1.5×10^{-4}	4.1×10^{-4}	5.1×10^{-4}
Complement deficiency genes (IUIS)	2 / 0.86	0.21	0.22	0.24
Rhythmic expression				
Rhythmic in ≥3 mouse tissues	105 / 82.8	0.0014	0.13	0.066
Human day–night (GTEx)	44 / 42.0	0.38	0.68	0.61



B



Organ systems with FDR q < 0.05: 5 under the primary null (filled circles), 0 after matching null loci on gene constraint.